Sifter Career Intelligence: Evidence, Mapping and Guidance Methodology

Status: Integrated methodology for academic review and product explanation
Audience: academic reviewers, curriculum partners, product reviewers, data reviewers, and future maintainers

1. What this document explains

Sifter connects two decisions that are often kept separate:

1. **Admission and programme decisions:** which programmes a learner may investigate from O'Level subjects, Unified Tertiary Matriculation Examination subjects, or both, and which universities offer the selected programme.
2. **Career understanding and planning:** what the programme teaches, which professional domains it connects to, which specialisations and roles exist inside those domains, and what skills, competencies, tools, credentials and evidence a learner may need to build.

The complete relationship is:

```
learner type and study stage
  -> O'Level and/or Unified Tertiary Matriculation Examination evidence
  -> national Joint Admissions and Matriculation Board programme identity
  -> university offering and university-specific requirements
  -> Nigerian Universities Commission curriculum evidence
  -> programme capabilities and skills taught
  -> related professional domains
  -> subgroups or a direct domain-to-role route
  -> roles and progression stage
  -> competencies, skills, tools, certifications, resources and opportunities
  -> a saved, explainable guidance plan
```

This is not a claim that a subject combination guarantees admission, that a programme guarantees a job, or that a domain relationship guarantees competence. It is an evidence-based guidance model that explains what each relationship means and what remains to be built.

2. The central evidence principle

Every important statement has three separate parts:

1. **Source fact:** what a document, framework, regulator, curriculum, or market source actually says.
2. **Derived relationship:** the documented rule used to connect that fact to another layer of the product.
3. **User explanation:** the plain-language conclusion shown to the learner.

For example:

Layer	Example
Source fact	A Nigerian curriculum document lists courses and learning outcomes involving programming, algorithms, and software development.

Layer	Example
Derived relationship	Those anchors overlap materially with the foundation of the Software Engineering domain.
User explanation	“Your programme teaches a foundation closely related to this domain.”

The existence of a source does not automatically create a relationship. The relationship must satisfy a stated rule, and the evidence and rule must remain visible to reviewers.

3. The source architecture

No single source provides the whole product. Each source is used for the kind of question it is qualified to answer.

3.1 Nigerian admission and curriculum sources

Source	What it contributes	What it does not prove
Joint Admissions and Matriculation Board (JAMB) and Integrated Brochure and Syllabus System (IBASS)	National programme names, Unified Tertiary Matriculation Examination combinations, O'Level requirements, direct-entry wording, and programme identity	That a learner will be admitted, a university-specific exception, or a career outcome
National Universities Commission (NUC) Core Curriculum and Minimum Academic Standards (CCMAS)	Programme purpose, philosophy, objectives, courses, learning outcomes, course content, and graduate capabilities	That every university teaches every course identically or that a programme guarantees a job
Nigerian regulators and professional bodies	Nigerian terminology, professional boundaries, registration, licensing, examinations, credential routes, and regulated practice	A universal requirement unless the particular regulator or law states it
Nigerian universities and official offering information	Which university offers a programme, location, university-specific notes, and local admission conditions	A national rule that applies to every university
Nigerian vacancy and labour sources	Dated Nigerian role titles and local demand signals where the checked source directly contains them	Complete Nigerian demand, future demand, or proof that a global role is absent locally

Admission and curriculum sources are deliberately kept separate from international occupation sources. A curriculum document explains what a programme teaches; an occupation framework explains what work is performed.

3.2 International classification and occupational sources

Source	Full name and contribution	Boundary
ISIC	International Standard Industrial Classification of All Economic Activities. It surveys industries and economic activities so major areas of work are not overlooked.	It is not an occupation, subject, skills, or programme-suitability catalogue.

Source	Full name and contribution	Boundary
ISCO-08	International Standard Classification of Occupations, maintained by the International Labour Organization. It supplies occupational groups, work functions, and unit-group anchors.	It does not name all Sifter domains or establish Nigerian demand.
ESCO	European Skills, Competences, Qualifications and Occupations. It supplies normalized occupation labels, alternative labels, descriptions, identifiers, and occupation-skill relationships.	It is international vocabulary, not Nigerian licensing or vacancy evidence.
O*NET	Occupational Information Network. It supplies tasks, work activities, skills, knowledge, abilities, technology and software signals, preparation, job zones, and work context.	It is primarily a United States occupational reference and is used as a global proxy.
ILO	International Labour Organization occupational and labour information. It supports interpretation of ISCO and international employment context.	It does not replace Nigerian regulators or local market evidence.
WEF	World Economic Forum Future of Jobs research. It supplies trend and skills-on-the-rise signals.	It is not a complete occupation dictionary or a Nigerian vacancy source.
Specialist frameworks	Frameworks such as SFIA, NICE, ASCM, CIPS, PMI, health, engineering, creative, immersive technology, quantum, robotics, and esports frameworks.	They enrich a professional area but do not automatically create a Nigerian programme link.

3.3 Public source references

The principal public references used for the methodology are:

- JAMB and IBASS programme information
- National Universities Commission
- International Standard Industrial Classification
- International Standard Classification of Occupations
- ESCO classification
- O*NET database
- International Labour Organization
- World Economic Forum Future of Jobs research
- National Digital Economy Policy and Strategy resources
- Council for the Regulation of Engineering in Nigeria
- Nursing and Midwifery Council of Nigeria
- Computer Professionals Registration Council of Nigeria
- National Board for Technical Education
- National Bureau of Statistics labour information
- Jobberman Nigeria and MyJobMag
- Quantum Technology Competence Framework
- JFF immersive learning and extended reality pathways
- British Esports career pathways

A link identifies the public source. The evidence record must additionally retain the source version, relevant page or section, extracted fact, and decision that the fact supports. A source registry or URL alone is not evidence that a particular claim was established.

4. Subject matching: the admission starting point

4.1 Subject identity and layers

The subject catalogue contains separate examination layers. It is not one large list in which every subject can be used for every examination.

- **O'Level subjects** are subjects from the learner's secondary-school result.
- **Unified Tertiary Matriculation Examination subjects** are selected for the national entrance examination.
- Technical and trade subjects remain O'Level choices unless a checked source explicitly makes them valid for the entrance examination.
- Aliases are normalized to one selectable subject identity. A source variation such as “General Wood Work” can point to “Woodwork” without creating a second selectable subject.
- Stream labels influence the order of choices; they do not hide a valid subject or change the underlying national requirement.

Normalization prevents duplicate subjects, false mismatches, and inconsistent results caused by matching printed phrases directly.

4.2 O'Level-only matching

In O'Level-only mode, the learner supplies the configured number of normalized O'Level subjects. The matcher uses only O'Level requirement records.

It evaluates separately:

1. named required subjects;
2. one-of or any-of alternatives;
3. the minimum number of credits or subjects; and
4. source-backed grade, sitting, or other conditions.

It does not use a UTME rule to fill a missing O'Level subject. The result means that the learner can investigate programmes whose checked O'Level requirements match the supplied evidence. It is guidance, not an admission guarantee.

4.3 UTME-only matching

In UTME-only mode, the learner supplies the required four normalized UTME subjects, including Use of English where the national rule requires it. The matcher uses only typed UTME option records.

Required subjects and alternatives are evaluated independently. A subject appearing only in an unparsed source paragraph is not treated as an exact match. The result means that the checked national UTME subject rule matches; it does not establish O'Level eligibility or university admission.

4.4 Combined O'Level and UTME matching

Combined mode does not flatten the two examinations into one list. It performs two independent matches and joins their results through the same canonical JAMB programme identity:

```
O'Level subjects -> O'Level requirement -> canonical JAMB programme ID
UTME subjects   -> UTME requirement   -> canonical JAMB programme ID
                                                  |
                                      intersection of programme IDs
```

This distinguishes:

- a programme that matches both layers;
- a programme whose UTME subjects match but whose O'Level evidence has a gap;
- a programme whose O'Level evidence matches but whose UTME combination does not; and
- a programme whose source rule is too ambiguous to parse safely.

The user sees the missing or uncertain layer rather than a misleading combined approval.

4.5 Example of the three modes

Assume a checked programme rule requires:

- UTME: English, Chemistry, Biology, and one of Mathematics or Physics;
- O'Level: English, Biology, Mathematics, Chemistry, and one of Geography, Health Science, Physics, or Statistics.

A learner selecting English, Chemistry, Biology, and Physics can satisfy the UTME rule because Physics satisfies the alternative group. A learner with the required O'Level credits can satisfy the O'Level rule independently. Combined mode retains the programme only when the same canonical programme identity is present in both results, and still displays any O'Level gap.

The example demonstrates the logic; it is not a claim that every programme uses the same subjects.

5. Programme identity and university connection

5.1 One national programme identity

The JAMB programme name is the national programme identity used to connect subject matching, university offerings, curriculum evidence, and later career guidance. A university does not become a new programme identity merely because it has a different admission note.

```
canonical JAMB programme
  -> universities that offer that JAMB programme
  -> address and location
  -> that university's admission requirements and notes
  -> the programme's curriculum and career explanation
```

5.2 University-specific requirements

After the national programme match, the learner sees the universities offering that programme. The programme preview comes first so the learner can understand what the programme is before choosing a university.

The university layer adds:

- university name and location;
- confirmation that it offers the selected JAMB programme;
- university-specific UTME and O'Level requirements;

- direct-entry or sitting conditions where applicable; and
- a note that the requirement applies to that university offering only.

The programme is not redefined for each university. Differences in requirements are attached to the university-programme pair. Raw exceptions remain visibly conditional until safely interpreted.

5.3 User-stage behavior

- **SS1 and SS2 learners:** begin with O'Level-oriented exploration. They can understand programme previews and see universities offering a programme without being forced to provide a UTME combination before they are ready.
- **SS3 and JAMB candidates:** can use O'Level-only, UTME-only, or combined matching according to what they want to check. They see the programme preview before selecting a university and then see that university's details.
- **Undergraduates:** normally enter through their university and programme; the admission subject matcher is not imposed as their default route.
- **Administrators guiding a student:** first select the student's context and applicable stream, then choose whether to inspect the full journey, the admission layer, the university layer, or the career-guidance layer. The administrator's own profile must not silently replace the student's context.

6. What a programme means in the product

The programme is not merely a title and not a list of possible jobs. Its overview answers:

1. What is this programme about?
2. What does the school teach through it?
3. What are the main curriculum themes?
4. What do important learning outcomes mean in ordinary language?
5. What should a graduate understand or be able to do by graduation?
6. Which transferable capabilities may remain useful beyond one occupation?
7. Which skills are explicitly taught or supported by curriculum evidence?

The concise programme preview remains simple. It is not the place to display every skill, domain, or role. The detailed programme view can expand into curriculum themes, skills taught, outcomes, transferable capabilities, and the evidence behind programme-to-domain connections.

The product uses **skills taught** as the simple learner-facing summary. It does not need a separate course-by-course capability catalogue in the preview. The underlying course and outcome evidence remains available for explanation and verification when the learner chooses to view more.

7. How curriculum is connected to domains

7.1 Why curriculum is the starting evidence

The programme-to-domain relationship is about what the programme prepares a learner to begin understanding, not merely what occupations exist in the world. Therefore Nigerian CCMAS evidence is the primary source for this link.

Reviewers inspect the available combination of:

- programme purpose and philosophy;

- named courses and course content;
- stated learning outcomes;
- graduate capabilities;
- transferable capabilities; and
- domain-specific concepts or prerequisites.

The evidence is compared with the independently defined domain foundation, occupational work outputs, entry routes, and boundaries. A word match is never enough. “Data” in a course title does not by itself prove a Data and Analytics relationship.

7.2 The domain is the comparison target

The domain side contributes its own evidence:

- the professional problem the domain addresses;
- common work outputs;
- the portable foundation needed across roles;
- common tools and methods;
- subgroups and role families;
- regulated boundaries where relevant; and
- Nigerian and global context.

The mapping asks two questions together:

1. **Curriculum question:** what did the programme actually teach or say a graduate should be able to do?
2. **Domain question:** does that preparation overlap with the target domain's foundation and work identity, and how much additional preparation is needed?

This explains the link from both directions instead of relying on a curriculum-only similarity score.

7.3 Direct, functional, and adjacent relationships

These labels describe the strength and kind of relationship. They do not mean that one student is already qualified.

Direct

A Direct relationship exists when the programme's purpose, curriculum, or outcomes support a substantial part of the target domain's foundation and occupational direction. It requires at least two specific, source-backed curriculum anchors, such as named courses, objectives, or learning outcomes.

The domain link must also be independently supported by occupational or professional evidence. Two similar course titles without a meaningful domain foundation are insufficient.

Learner explanation:

> Your programme teaches a foundation closely related to this domain.

Functional

A Functional relationship exists when the programme teaches capabilities that are materially useful in the target domain but does not establish the complete domain foundation. It also requires at least two specific, source-

backed curriculum anchors.

The capabilities are relevant; the learner must add important domain-specific knowledge, practice, tools, or evidence before specialising.

Learner explanation:

> Your programme gives you useful preparation for this domain, but you will need to build additional domain-specific capability.

Adjacent

An Adjacent relationship is a plausible pivot. The programme supplies transferable capabilities plus at least one specific conceptual prerequisite related to the target domain, but not enough of the target occupational foundation to call the relationship Direct or Functional.

The bridge is part of the relationship. It must say what the learner would need to learn before specialising.

Learner explanation:

> This is a possible pivot. Your current foundation helps, but you need this bridge before specialising.

7.4 The meaning of strong evidence

“Strong” has one consistent meaning across the relationship classes: specific, source-backed curriculum anchors support the relationship, and the domain side has an independently supported identity and foundation.

- Direct: at least two meaningful curriculum anchors overlap with the domain foundation and occupational direction.
- Functional: at least two meaningful anchors support useful capability, while important domain preparation is missing.
- Adjacent: at least one domain-specific conceptual prerequisite plus relevant transferable-capability evidence supports a bridge, but not a full foundation.

These are evidence gates, not probabilities of employment, admission, or success.

7.5 Programme-to-domain example

Consider a Computer Science programme as an illustrative example. A checked curriculum may include programming, algorithms, software design, databases, systems, and computing ethics. Domain evidence for Software Engineering may describe software systems, development practices, testing, and delivery. The overlap in purpose, foundation, and work outputs can support a Direct link.

The same programme may connect Functionally to Data and Analytics when it teaches programming, databases, and quantitative reasoning but does not cover the full analytics, experimentation, or business-intelligence foundation. A Cybersecurity relationship may be Direct, Functional, or Adjacent depending on the actual security curriculum anchors; the title alone cannot decide it.

Every decision must point to the precise curriculum anchors and domain evidence used for the comparison.

8. How domains were identified, named, and bounded

8.1 ISIC is the completeness survey, not the domain-name generator

The International Standard Industrial Classification begins with economic activities: agriculture, manufacturing, information and communication, finance, health, education, arts, and others. It helps ask whether an important area of work has been overlooked.

It does not directly produce the final learner-facing domain list. One economic section can contain several professional worlds with different foundations, entry routes, outputs, and daily practice. Information and communication, for example, can contain software engineering, cybersecurity, data, cloud infrastructure, product management, and digital media.

8.2 The defensible domain derivation process

For every relevant ISIC section and division, the review process is:

1. identify the economic activity and people performing professional work within it;
2. collect occupational families describing that work from ISCO, ESCO, O*NET, and relevant specialist frameworks;
3. group occupations sharing a meaningful professional foundation, work problem, output, entry route, employer context, and daily practice;
4. split groups when combining them would hide materially different learning foundations, regulated boundaries, or work outputs;
5. check that the group can support a coherent domain-to-subgroup-to-role pathway;
6. choose a name that practitioners and learners can understand; and
7. record source anchors, boundary decision, label class, and known limits.

The result is a source-traceable synthesis, not a claim that an international classification uses the exact same names.

8.3 Semantic and thematic umbrella method

An umbrella label is accepted through both semantic and thematic review.

Semantic review asks whether the candidate name has the same meaning as the source occupations and whether a learner would reasonably understand it to include those occupations.

Thematic review groups repeated evidence around:

- the professional problem being solved;
- the outputs delivered;
- knowledge and skills used;
- tools and methods used;
- work setting and employers;
- entry and progression routes; and
- legal or professional boundaries.

The process is:

1. collect exact source labels and descriptions;
2. extract repeated themes and work outputs;
3. compare candidate umbrella meanings with those themes;
4. test whether the label hides a materially different occupation;
5. validate the umbrella against occupational and specialist frameworks; and
6. explain the label in ordinary language without presenting it as an official classification title.

This avoids copying statistical language that students do not understand and inventing attractive labels with no occupational basis.

8.4 Three label classes

Each domain identity should state which kind of label it uses:

1. **Exact source terminology:** copied from a recognized source.
2. **Source-supported umbrella:** a plain-language grouping of multiple source occupations or activities sharing documented themes and boundaries.
3. **Curated addition:** added to close a documented coverage gap, with specialist evidence and a clear statement that it is a Sifter taxonomy addition rather than a verbatim official label.

The current master contains the original 166-domain library plus later, explicitly documented additions for Extended Reality and Immersive Technology and for Esports and the Creator Economy. The additions are not retroactively described as if they were in the original ISIC-derived library.

8.5 What makes the naming defensible

The defensible claim is not “ISIC named every Sifter domain.” It is:

> ISIC was used to survey industry coverage; occupational and professional sources identified the work families inside that coverage; and documented semantic and thematic criteria decided when those families formed separate learner-facing domains.

The source-backed name, source-supported boundary, and editorial decision must be recorded together. An academic reviewer can then challenge one boundary without treating the whole taxonomy as unsupported.

9. How subgroups were identified and named

A subgroup is a meaningful specialization inside a domain. It is not merely a smaller list created to fill a screen.

The subgroup boundary is supported by some combination of:

- ISCO sub-major groups or related occupational families;
- ESCO occupation groupings and shared skills;
- O*NET related occupations, tasks, and work context;
- specialist competency frameworks;
- Nigerian occupational standards and professional bodies; and
- the domain's shared foundation and distinct work outputs.

The review asks:

1. What work focus makes this subgroup different from the rest of the domain?
2. Which skills, methods, tools, or knowledge are shared inside it?
3. Which roles genuinely belong inside it?
4. What does it add beyond the domain foundation?
5. Can the boundary be explained without claiming it is an official government classification?

Subgroup names may be exact source terms, source-supported umbrellas, or documented curated labels. A subgroup must not be created merely because a role needs somewhere to be placed. If no defensible subgroup

exists, the product uses the honest route:

```
programme -> domain -> role
```

rather than fabricating a subgroup.

10. How roles were identified and connected

10.1 Role identity

Roles are connected to a domain or subgroup through their work identity, not a title that merely contains a related word. Principal role anchors are:

- exact ISCO-08 unit groups and occupational definitions;
- exact ESCO occupation labels, alternative labels, descriptions, and stable identifiers;
- O*NET occupation profiles, tasks, work activities, preparation, and tools;
- specialist competency and occupational frameworks; and
- Nigerian regulators, professional bodies, standards, and checked local role sources.

Where a role has several valid names, aliases are normalized to one role identity while source labels remain available. Where a source describes a role but no exact occupational code or profile can be established, the record says so. A source-role-only record is not automatically false; it has lower identity precision than an exact coded occupation.

10.2 Role connection to subgroup and domain

The relationship is supported when the role's principal work output, tasks, skills, tools, and work context fit the subgroup boundary and domain foundation. A role may belong to more than one domain only when its work identity and evidence justify each connection.

```
source occupational profile
  -> normalized role identity
  -> subgroup membership, where supported
  -> domain membership and foundation
  -> competencies, skills, tools, credentials, resources and evidence
```

A role is not a live Opportunity. **Opportunities** is the employment surface and appears only when a verified opportunity record exists.

10.3 Entry, mid, and senior progression

Roles are presented in preparation order so learners know what can be investigated or built first.

- **Entry:** a role reasonably explored after foundational study and initial evidence-building.
- **Mid:** a role normally expecting meaningful experience, deeper specialization, or broader independent responsibility.
- **Senior or lead:** a role involving advanced judgment, strategic ownership, supervision, complex delivery, or professional authority.

The tier is derived from preparation requirements, occupational skill level, job-zone or experience evidence, scope of responsibility, and reviewed role family rules. A senior-sounding title is not enough to make a role senior, and an entry label is not a promise of employment.

11. Competencies, skills, tools, and certifications

These terms are deliberately separate.

Term	Meaning in Sifter	Typical evidence
Competency	An integrated ability to apply knowledge, skills, judgment, and behavior to a work situation	Curriculum outcomes, professional standards, competency frameworks, occupational profiles
Skill	A specific learned ability, such as data analysis, programming, writing, or equipment operation	ESCO skills, O*NET skills and tasks, curriculum outcomes, specialist frameworks
Tool	Software, equipment, platform, method, or technology used to perform work	O*NET technology signals, role profiles, specialist frameworks, Nigerian sector evidence
Certification or registration	A credential, examination, membership, registration, or licence	Nigerian regulator, professional body, official credential provider, or clearly labelled employer preference

A source saying that a role uses a tool does not prove that the tool is legally required. A course outcome mentioning a skill does not prove every graduate has mastered it. A certification resource does not prove that the credential is mandatory in Nigeria.

11.1 How the inventory is populated

- ESCO occupation-skill relationships and descriptions supply normalized global skills and occupation connections.
- O*NET supplies tasks, knowledge, abilities, work activities, software, tools, preparation, and context.
- Specialist frameworks fill professional areas that broad classifications do not describe finely enough.
- CCMAS outcomes and content explain which capabilities the programme teaches or introduces.
- Nigerian standards, regulators, bodies, and local sources add local terminology, regulated preparation, credentials, and context.

No source silently changes category. A tool is not recast as a skill, a global skill is not recast as Nigerian demand, and a recommended course is not recast as a licence.

11.2 Priority and progressive disclosure

The product shows all available source-backed information but makes the next useful action clear. The rule is **complete information, selective progression**.

At domain level:

1. verified statutory or practice requirements, where applicable;
2. portable foundation competencies and skills;
3. tools used broadly across the domain;
4. preparation that opens the selected subgroup; and
5. specialist, advanced, optional, or context-dependent items.

At subgroup level:

1. the domain foundation that remains useful;
2. capabilities that close the subgroup gap;
3. tools used across that subgroup;
4. role-specific preparation; and
5. optional or advanced items.

At role level:

1. verified regulated preparation, where applicable;
2. essential skills, competencies, and tools for the work;
3. evidence-building and employability preparation;
4. Nigerian, remote, and global evidence with its scope stated; and
5. optional, advanced, or future-pivot preparation.

The interface shows the starting set first and keeps the complete inventory under **See all in this domain**, **See all in this subgroup**, or **See all for this role**. The full list remains inside the context that explains why each item is relevant.

Student-facing labels include:

- **Learn first** - suggested starting foundation, not a universal checklist.
- **Build next** - broadly useful preparation after the foundation.
- **Needed for this route** - supported by the selected subgroup or role evidence.
- **Recommended for employability** - useful but not universal.
- **Useful for a specialisation** - relevant after a path is chosen.
- **Optional or advanced** - useful later or in specific contexts.
- **Priority not yet verified** - present in the inventory but not supported strongly enough to call required or optional.

When evidence is incomplete, the product says **may need** rather than **need**.

12. Nigerian, remote, and global evidence

These are evidence lenses, not one blended score.

- **Nigerian context:** the field, practice, terminology, body, regulator, or institution is evidenced in Nigeria.
- **Nigerian market signal:** a checked Nigerian source directly contains the role, tool, competency, or demand signal.
- **Remote:** the source directly supports work accessible remotely; it is not inferred merely because a role is digital.
- **Global:** international standards, occupational profiles, or employers support the explanation.

The product must not say that a global occupation is in Nigerian demand merely because ESCO or O*NET contains it. A lack of a checked Nigerian vacancy must not erase a portable skill or global pathway. Local regulatory requirements receive priority where they apply, and the evidence scope is shown beside the explanation.

13. How the guided experience uses the mappings

The learner does not receive a flat export of all joins. The product tells a progressive story:

```
programme preview
  -> university options and university-specific requirements
  -> programme overview and skills taught
  -> why the programme connects to these domains
  -> direct domains first
  -> functional domains
  -> adjacent domains with an explicit bridge
  -> choose one primary domain and optional secondary domains
  -> learn the domain foundation
  -> choose a subgroup, where supported
  -> see what the subgroup adds
  -> choose an entry role first, then understand mid and senior progression
  -> review skills, competencies, tools, certifications, resources and evidence
  -> save the route and continue learning
```

The programme-to-domain screen explains the relationship before asking the learner to choose a route. A secondary or adjacent domain is available for comparison or pivot planning, but is not presented as if the learner has the same preparation as for the primary route.

Core guidance wording:

> You already have partial preparation from your programme and primary domain. These are the additional skills you need.

For a functional or adjacent route:

> This is a possible pivot. Your current foundation helps, but you need this bridge before specialising.

If evidence is incomplete:

> You may need these additional skills. The available evidence does not yet establish them as universal requirements.

14. What the learner should understand at every level

Programme

- what the programme is about;
- what the school teaches, summarized as skills taught;
- what important curriculum outcomes mean;
- what a graduate should understand or be able to do;
- which capabilities transfer beyond one occupation; and
- why the programme connects to each domain category.

Domain

- what the professional world does;
- what work outputs it produces;
- the portable foundation useful across roles;
- what to learn first;
- tools and competencies used across the domain;
- Nigerian, remote, and global evidence;

- regulated boundaries and credential context; and
- all other source-backed items under See all.

Subgroup

- what specialization it represents;
- how it differs from other parts of the domain;
- what the domain foundation already covers;
- what it adds beyond that foundation;
- its skills, competencies, tools, credentials, resources, and roles; and
- the bridge from the domain to the specialization.

Role

- what the work involves;
- expected outputs;
- what entry, mid, or senior positioning means;
- essential versus optional preparation;
- tools, skills, competencies, certifications, and proof of ability;
- Nigerian and global evidence with its limits; and
- related roles or possible pivots.

Each explanation can link to the original source. The link should be introduced with a prompt such as **Open the source explanation** or **See how this is defined**, so the learner understands why the link is offered.

15. Worked end-to-end examples

Example A: O'Level to programme to university

1. A learner enters O'Level subjects.
2. The matcher evaluates only O'Level rules and identifies programme names that satisfy the checked national requirements.
3. The learner opens a programme preview and learns what the programme is about and what skills are taught.
4. The learner sees all universities offering that JAMB programme.
5. After choosing a university, the learner sees its address, requirements, and notes.
6. The programme overview explains the curriculum and connected domains.

The university step does not rewrite the programme identity; it supplies local offering conditions.

Example B: UTME to programme to career guidance

1. A JAMB candidate enters four UTME subjects.
2. The matcher evaluates the typed UTME rule, including required and alternative subjects.
3. The candidate sees programme previews before choosing a university.
4. The selected university adds its requirements and exceptions.
5. CCMAS evidence explains what the programme teaches.
6. The curriculum is compared with domain foundations.

7. Direct domains appear first, functional domains next, and adjacent domains with a bridge explanation.

Example C: Programme to domain to subgroup to role

1. Curriculum evidence supports a Direct connection to Software Engineering.
2. The learner chooses Software Engineering as the primary domain.
3. The domain view starts with shared programming, systems, testing, and delivery foundations while keeping other items under See all.
4. The learner chooses a supported subgroup such as application development or platform engineering when the evidence supports that boundary.
5. The subgroup view shows what is shared with the domain and what must be added.
6. Entry roles appear before mid and senior progression roles.
7. The selected role explains work outputs, tools, skills, competencies, credentials, evidence, resources, and Nigerian or global limitations.

If a domain has roles but no defensible subgroup taxonomy, step 4 is omitted and the learner goes directly from domain to role.

16. Evidence states and honest limitations

The system distinguishes:

- **Source-backed:** the source directly supports the stated fact.
- **Source-supported umbrella:** multiple source records are grouped under a documented, understandable label.
- **Nigerian context:** the field or practice is evidenced in Nigeria.
- **Nigerian market signal:** a checked local source directly supports the market observation.
- **Global evidence:** an international source supports the occupation, skill, task, or trend.
- **Bridge required:** the relationship is plausible, but additional domain-specific preparation is needed.
- **No direct Nigerian counterpart found:** checked Nigerian sources did not establish an exact local counterpart; this is not proof that the role does not exist in Nigeria.
- **Review required:** the source or identity is not sufficiently resolved for a stronger claim.

An unresolved occupational code is a provenance and code-quality issue. The original value remains visible until its meaning is matched to a valid canonical code. It must not be silently replaced with a guessed code. This limits the strength of the occupational-code claim but does not automatically invalidate an otherwise supported domain or role explanation.

A missing Nigerian vacancy, tool, or certification record is not silently converted into an optional item. The product explains the gap and preserves global or portable information when appropriate.

17. What can be defended academically

The strongest defensible description is:

> Sifter is a source-traceable, human-reviewed synthesis. It uses Nigerian admission and curriculum documents to establish learner entry and programme context; uses the International Standard Industrial Classification to survey industry coverage; uses the International Standard Classification of Occupations and the European Skills, Competences, Qualifications and Occupations classification to structure and

> normalize occupational identity; uses the Occupational Information Network, International Labour Organization, World Economic Forum, and specialist frameworks to enrich work, skills, tools, preparation, and trends; and uses Nigerian regulators, professional bodies, universities, and market sources to add local context. Learner-facing names and links are derived through explicit semantic, thematic, occupational, curriculum, and boundary rules rather than copied from one classification or generated without evidence.

This claim is appropriately modest. It does not say every source agrees, every role has a Nigerian counterpart, every credential is mandatory, or a curriculum-to-domain link guarantees employment.

18. Review questions for every mapping

A reviewer should be able to ask these questions about any subject rule, programme, domain, subgroup, role, competency, tool, or credential:

1. What exactly is the identity of this record?
2. Which source established that identity?
3. Which page, section, URL, or source passage supports the claim?
4. Is the label copied, an evidence-supported umbrella, or curated?
5. What rule created the relationship to the next layer?
6. What evidence supports the other side of the relationship?
7. Is this Nigerian context, Nigerian market evidence, remote evidence, or global evidence?
8. What does the learner already have, and what bridge is still required?
9. Is the item required, essential for the route, recommended, optional, or not yet verified?
10. What should the learner see if the evidence is missing, disputed, or unavailable?

If these questions cannot be answered, the relationship should be weakened, labelled under review, or excluded from a stronger user-facing claim.

19. Final summary

The product's logic is not:

```
subjects -> similar programme name -> list of jobs
```

It is:

```
subject evidence
  -> checked national programme requirement
  -> canonical programme identity
  -> university-specific offering context
  -> Nigerian curriculum and skills taught
  -> independently evidenced domain identity and foundation
  -> evidence-based Direct, Functional, or Adjacent relationship
  -> supported subgroup or honest direct domain-to-role route
  -> normalized role and progression stage
  -> prioritized competencies, skills, tools, credentials and resources
  -> Nigerian/global evidence and explicit limitations
  -> guided, saved, explainable learner plan
```

The learner sees all available information, but the product explains what to start with, why it matters, what can wait, and what evidence supports the connection. That is the basis on which the mappings can be reviewed, challenged, improved, and responsibly used before implementation or release. +

20. Teaching examples at every layer

The following examples show how the method works. They are explanatory examples of the rules; a production mapping is accepted only when its own source passage and evidence record satisfy the same rule.

20.1 Example: O'Level matching

Suppose a learner has English Language, Mathematics, Biology, Chemistry, and Physics at O'Level.

The matcher does not ask whether those subjects “sound related” to a programme. It retrieves the programme's checked O'Level rule and evaluates:

- the named required subjects;
- any alternative group;
- the required number of subjects or credits; and
- any grade, sitting, or condition printed in the source.

If a programme requires English, Mathematics, Biology, Chemistry, and one of Physics or Geography, this learner satisfies the subject pattern. The product may show the programme as an O'Level match, but it must not say that admission is guaranteed.

If the learner has English, Biology, Chemistry, and Physics but no Mathematics, the product should show the missing requirement. It must not use a UTME Mathematics rule to hide the O'Level gap.

20.2 Example: UTME-only matching

Suppose a checked UTME rule requires English, Chemistry, Biology, and one of Mathematics or Physics.

A candidate selecting English, Chemistry, Biology, and Physics satisfies the UTME subject combination. The result is a UTME programme match only. It does not certify the candidate's O'Level result, university-specific requirement, or admission outcome.

This distinction matters because a programme can be a valid UTME match while the candidate still has an O'Level deficiency.

20.3 Example: combined matching

A candidate's O'Level result matches a programme, and the candidate's UTME combination also matches it. The system joins the two results by the same canonical JAMB programme identity.

It does not join by a similar title. For example, a programme with a longer university display name must still resolve to the same national programme identity before the two examination layers are intersected.

The result can then say:

- “Your O'Level and UTME evidence match the checked national requirements”; or
- “Your UTME subjects match, but an O'Level requirement still needs attention.”

20.4 Example: programme preview and university choice

A learner first sees:

- the JAMB programme name;
- a short explanation of what the programme is about; and
- the concise “skills taught” summary.

Only after understanding that preview does the learner see the universities offering that programme. After selecting a university, the learner sees the university's location, requirements, and university-specific notes.

If University A requires a particular O'Level grade and University B does not, that difference belongs to the university-programme offering. It does not create two unrelated programme identities.

20.5 Example: programme curriculum explanation

For an Accounting programme, the curriculum may contain financial reporting, cost accounting, taxation, auditing, economics, business law, and information systems.

The product can explain these in plain language:

- financial reporting means preparing and interpreting records of an organisation's financial activity;
- auditing means examining records and controls to assess whether they are reliable and compliant;
- taxation means understanding how tax obligations are calculated and managed;
- information systems means using structured systems and data to support financial work.

The source fact comes from the curriculum. The plain-language sentence is an explanation of that fact, not a new claim about the learner's competence.

20.6 Example: a curriculum-to-domain Direct link

A Computer Science curriculum that includes programming, algorithms, software design, databases, systems, testing, and computing ethics may support a Direct relationship to Software Engineering.

The decision is not caused by the words “computer” or “software” alone. It is caused by the overlap between:

- the programme's documented learning anchors;
- the Software Engineering domain foundation;
- the domain's work outputs, such as designing, building, testing, and maintaining software systems; and
- occupational evidence describing that work.

The learner sees:

> Your programme teaches a foundation closely related to this domain.

20.7 Example: a curriculum-to-domain Functional link

An Economics curriculum may contain quantitative analysis, statistics, research, markets, and decision-making. Those capabilities can support a Functional relationship to Data and Analytics when the programme does not provide the full technical foundation of data engineering, programming, experimentation, or production analytics.

The learner sees:

> Your programme gives you useful preparation for this domain, but you will need to build additional domain-specific capability.

The missing preparation is part of the mapping. It may include programming, data tooling, data modelling, or portfolio evidence, but each item must have its own supporting source and evidence state.

20.8 Example: a curriculum-to-domain Adjacent link

A Biology programme may teach scientific reasoning, laboratory practice, measurement, and interpretation of evidence. Those capabilities can support an Adjacent relationship to Health Data and Medical Informatics when the programme contains a relevant health or data prerequisite but does not teach the full informatics foundation.

The learner sees:

> This is a possible pivot. Your current foundation helps, but you need this bridge before specialising.

The bridge might include health-information standards, data management, statistics, or software practice, depending on the actual evidence. The bridge cannot be invented from the domain title.

20.9 Example: domain identity and definition

For a Finance domain, the identity review may find:

- ISIC financial and insurance activities as the industry context;
- ISCO occupations for financial analysts, advisers, risk specialists, and related work;
- ESCO occupation names, descriptions, and skills;
- O*NET tasks, preparation, tools, and work context;
- CFA Institute or Global Association of Risk Professionals material for specialist boundaries; and
- Nigerian Central Bank, Securities and Exchange Commission, Institute of Chartered Accountants of Nigeria, Chartered Institute of Bankers of Nigeria, and local market sources for Nigerian context.

The learner-facing name “Finance and Investment” can be a source-supported umbrella when those sources describe a coherent work family. It is not claimed to be the official name of ISIC or ISCO.

For a Health domain, the same method may distinguish clinical practice, public health, health data, pharmacy, and mental health because their foundations, regulated boundaries, work outputs, and entry routes differ.

For Extended Reality and Immersive Technology, the identity is a documented specialist addition supported by immersive-learning, standards, and occupational sources. It must not be described as though ISIC originally contained that exact learner-facing label.

20.10 Example: subgroup derivation

Inside a Data and Analytics domain, a subgroup might represent data engineering when its roles share data pipelines, storage, modelling, quality, and platform operations.

Inside a Creative domain, a subgroup might represent technical theatre and production when its roles share live-event systems, staging, technical coordination, and production safety.

The subgroup is accepted only when:

- source occupational families support the work focus;
- the roles share a meaningful foundation;
- the subgroup adds something beyond the parent domain; and
- the boundary can be explained to a learner.

If a domain has credible roles but no such boundary, the product uses domain-to-role navigation. It does not invent a subgroup to make the screen look uniform.

20.11 Example: role identity

A role such as Data Analyst is connected through its work, not merely because the word “data” appears in its title. Reviewers compare:

- the exact or normalized occupational label;
- tasks and outputs;
- required or commonly used skills;
- tools and work context;
- preparation level; and
- its relationship to the subgroup and domain.

A role such as Medical Laboratory Scientist also requires local boundary review. International occupational evidence may explain laboratory work, but the Medical Laboratory Science Council of Nigeria determines which Nigerian registration or practice claims can be made.

A role such as Game Developer may be supported by game-development frameworks, O*NET or ESCO-related evidence, specialist industry sources, and a Nigerian vacancy or creative-industry source where available. A global role source does not become Nigerian demand simply because it is in the role library.

20.12 Example: skills, tools, and certifications

For a Procurement subgroup:

- procurement planning is a competency or integrated capability;
- supplier evaluation is a skill;
- an enterprise resource planning system is a tool;
- Chartered Institute of Procurement and Supply preparation is a credential route; and
- a Nigerian professional-body requirement, if explicitly stated, is local regulatory or professional evidence.

These should not be merged into one “job readiness” score. The learner needs to know what each item is, why it appears, and whether it is required, recommended, optional, or not yet verified.

For an Engineering role:

- engineering analysis may be a role competency;
- computer-aided design may be a skill or method;
- a design application may be a tool; and
- Council for the Regulation of Engineering in Nigeria registration may be a regulated professional requirement.

The product must explain that learning a tool is not the same as obtaining professional registration.

20.13 Example: priority inside a domain, subgroup, and role

For a Cybersecurity domain, the domain page may first show security principles, networking, operating systems, identity, and safe investigation. It keeps specialist penetration-testing tools, advanced cloud security, and niche certifications under See all unless the learner chooses that route.

After selecting a defensive-security subgroup, the page shows what the domain foundation already covers, then adds incident response, monitoring, detection, and evidence handling.

After selecting an entry security analyst role, the page shows role essentials and starter evidence before mid-level threat hunting and senior security architecture. This is progressive disclosure, not deletion of information.

20.14 Example: Nigerian and global evidence

For a finance role, a Nigerian banking or regulatory source can establish local terminology, local practice, or a regulatory boundary. CFA Institute material can explain an international investment pathway. A Jobberman listing can provide a dated Nigerian market signal for a role title.

Those three facts are not interchangeable:

- professional practice in Nigeria is not the same as a vacancy count;
- an international credential is not automatically a Nigerian licence; and
- one vacancy is not proof of universal demand.

The interface must show the evidence lens next to the explanation.

21. Complete public source reference register

This register names every source family currently included in the source inventory. It states why each source is relevant and links to its public location where one is available. A source being listed means it is an authorized source route; the evidence record must still say whether the document or extract was actually checked for the particular claim.

21.1 Admission and Nigerian curriculum

- **Joint Admissions and Matriculation Board:** national admission information, JAMB Frequently Asked Questions.
- **Integrated Brochure and Syllabus System:** programme names and admission subject rules, IBASS.
- **National Universities Commission:** university standards and CCMAS, NUC.
- **National Board for Technical Education:** technical curricula and National Occupational Standards, NBTE downloads and National Skills Qualification Framework resources.
- **National Bureau of Statistics:** Nigerian labour-force context, NBS microdata catalogue.
- **Nigerian National Artificial Intelligence Strategy:** checked national strategy document used for local artificial-intelligence context. Its public hosting must be recorded with the document version when a claim uses it; a registry entry alone is not sufficient evidence.
- **Nigerian university offering pages:** institution, programme, location, local requirements, and notes. The university's official page is the source for its own offering and is not treated as a national rule.

The admission input set includes the official JAMB brochure sections for Administration, Agriculture, Arts and Humanities, Education, Engineering and Environmental Technology, Law, Medical and Pharmaceutical Health Science, Sciences, Social and Management Science, and Specific Entry Requirements.

The curriculum input set includes the NUC CCMAS families for Administration and Management, Agriculture, Allied Health Sciences, Architecture, Arts, Basic Medical Sciences, Communication and Media Studies, Computing, Education, Engineering and Technology, Environmental Sciences, Law, Medicine and Dentistry, Pharmacy and Pharmaceutical Sciences, Sciences, Social Sciences, and Veterinary Medicine.

21.2 International classifications

- **United Nations International Standard Industrial Classification of All Economic Activities:** industry and activity completeness context, UN ISIC classification page.
- **United Nations ISIC explanatory notes:** activity definitions and revision interpretation, ISIC explanatory notes.
- **International Labour Organization International Standard Classification of Occupations:** occupational hierarchy and work-function anchors, ISCO-08.
- **International Labour Organization ISCO-08 Volume 1:** definitions and structure, ISCO-08 Volume 1.
- **European Skills, Competences, Qualifications and Occupations:** normalized occupations, skills, descriptions, and identifiers, ESCO use and classification resources.
- **Occupational Information Network:** tasks, skills, tools, preparation, and work context, O*NET database.
- **International Labour Organization:** international labour and occupational context, ILO.
- **World Economic Forum Future of Jobs 2025:** future work and skills trends, WEF report.

ISIC is the industry-context survey. ISCO, ESCO, and O*NET provide occupational structure and enrichment. None of them alone generates the complete Sifter domain list or proves Nigerian demand.

21.3 Global professional and specialist sources

- **SFIA 9:** digital and information-technology skills, SFIA.
- **NIST NICE Workforce Framework:** cybersecurity work and competency language, NICE Framework.
- **Association for Supply Chain Management:** supply-chain body of knowledge, ASCM.
- **Chartered Institute of Procurement and Supply:** procurement and supply standard, CIPS Global Standard.
- **Project Management Institute:** project-management standards, PMI standards.
- **World Health Organization:** health-workforce context, WHO health workforce.
- **Food and Agriculture Organization:** agriculture and food systems, FAO.
- **World Organisation for Animal Health:** animal-health standards, WOAH.
- **International Civil Aviation Organization and International Air Transport Association:** aviation standards, ICAO and IATA.
- **International Maritime Organization and Institute of Marine Engineering, Science and Technology:** maritime standards, IMO and IMarEST.
- **Royal Institution of Chartered Surveyors:** property and built-environment standards, RICS.
- **Society of Petroleum Engineers:** petroleum engineering competencies, SPE competency management.
- **International Atomic Energy Agency:** nuclear and radiation work, IAEA.
- **United Nations Educational, Scientific and Cultural Organization:** education, science, and culture context, UNESCO.
- **ScreenSkills:** creative-industry role profiles, ScreenSkills.
- **International Game Developers Association:** game-development professional resources, IGDA.
- **Philippine Skills Framework for Game Development:** game-development progression and skills, TESDA framework.
- **Khronos Group:** open standards relevant to immersive and graphics work, Khronos.
- **European Competence Framework for Quantum Technologies:** quantum technology competencies, Quantum Flagship framework.
- **Singapore National Arts Council technical theatre and production framework:** technical theatre work, Skills Framework.
- **Skills England Live Event Technician standard:** live-event work, Skills England.

- **Malaysian Qualifications Agency Performing Arts standards:** performing arts programme and role context, MQA standards.
- **Jobs for the Future immersive-learning and extended-reality pathways:** immersive careers, JFF XR pathways.
- **RoboticsCareer:** robotics competency pathways, RoboticsCareer.
- **Skills England Robotics Engineer maps:** robotics work and progression, Robotics Engineer and Advanced Robotics.
- **British Esports:** esports education and career pathways, British Esports.
- **Pearson BTEC Esports:** esports roles and programme pathways, Pearson esports specification.
- **Hitmarker:** esports and gaming employment vocabulary, Hitmarker.
- **International Federation of the Phonographic Industry:** music-industry context, IFPI.
- **World Intellectual Property Organization:** intellectual-property context, WIPO.
- **CFA Institute:** investment-profession preparation, CFA Institute.
- **Global Association of Risk Professionals:** risk-profession preparation, GARP.
- **Driver and Vehicle Standards Agency:** driver and rider training standard, DVSA standard.

21.4 Nigerian regulators, professional bodies, and sector sources

- **Jobberman Nigeria:** dated Nigerian vacancy and role-title signals, Jobberman.
- **MyJobMag Nigeria:** dated Nigerian vacancy and role-title signals, MyJobMag feeds.
- **Nigeria Computer Society:** Nigerian computing context, NCS.
- **Computer Professionals Registration Council of Nigeria:** computing professional context and examination material, CPN and CPN syllabus.
- **Council for Innovative Engineering and Emerging Engineering:** Nigerian emerging-engineering context, CIEEEN.
- **Federal Ministry of Health and Social Welfare:** Nigerian health-service context, hospital services.
- **Medical Laboratory Science Council of Nigeria:** laboratory-science practice and approved documents, MLSCN.
- **Council of Registered Builders of Nigeria:** construction standards and labour-market information, CORBON.
- **Quantity Surveyors Registration Board of Nigeria:** quantity-surveying registration, QSRBN.
- **Nigerian Nuclear Regulatory Authority:** nuclear and radiation regulation, NNRA.
- **Teachers Registration Council of Nigeria:** teaching registration, TRCN.
- **Nigerian Oil and Gas Industry Content Joint Qualification System:** local oil-and-gas competency and qualification context, NOGIC JQS.
- **Centre for Management Development:** Nigerian management development, CMD.
- **National Space Research and Development Agency:** Nigerian space and space-science context, NASRDA.
- **National Emergency Management Agency:** emergency-management context, NEMA.
- **Federal Ministry of Arts, Culture and Creative Economy:** Nigerian creative-economy context, FMaCCE.
- **Institute of Chartered Accountants of Nigeria:** Nigerian accounting preparation, ICAN syllabus.
- **Chartered Institute of Bankers of Nigeria:** Nigerian banking competencies, CIBN competency framework.
- **Nigeria Union of Journalists:** Nigerian journalism ethics, NUJ code of ethics.
- **Nigerian Library Association and Librarians Registration Council of Nigeria:** library and information-profession context, NLA.
- **Nigerian Council of Food Science and Technology:** food-science context, NICFOST.

- **National Institute for Hospitality and Tourism:** hospitality registration context, NIHOTOUR RIVAC.
- **Federal Ministry of Agriculture and Food Security:** Nigerian agriculture and food systems, FMAFS.
- **Nigerian Psychological Association:** psychology context, NPA.
- **Nigerian Fashion Council and Fashion Designers Association of Nigeria:** fashion context and registration, FADAN.
- **Nigeria Olympic Committee:** Nigerian sports context, NOC athletes commission.
- **National Primary Health Care Development Agency:** Nigerian public-health context, NPHCDA.
- **Nigeria Meteorological Agency:** meteorology context, NiMet.
- **Nigerian Upstream Regulatory Commission and Nigerian Midstream and Downstream Petroleum Regulatory Authority:** energy regulation, NUPRC and NMDPRA.
- **Nigerian Bar Association:** Nigerian legal context, NBA.
- **Chartered Institute of Purchasing and Supply Management of Nigeria:** Nigerian procurement context, CIPSMN.
- **Nigerian Academy of Science:** Nigerian research context, NAS.
- **National Communications Commission and National Information Technology Development Agency:** Nigerian telecommunications and digital ecosystem, NCC and NITDA.
- **Council for the Regulation of Engineering in Nigeria and Nigerian Society of Engineers:** Nigerian engineering practice, COREN and NSE.
- **Architects Registration Council of Nigeria and Surveyors Council of Nigeria:** built-environment professional boundaries, ARCON and SURCON.
- **Medical and Dental Council of Nigeria, Nursing and Midwifery Council of Nigeria, and Pharmacists Council of Nigeria:** Nigerian health-profession boundaries, MDCN, NMCN, and PCN.
- **Institute of Chartered Accountants of Nigeria, Securities and Exchange Commission, and Central Bank of Nigeria:** Nigerian finance and accounting context, ICAN, SEC, and CBN.
- **Chartered Institute of Personnel Management of Nigeria and Nigerian Institute of Management:** Nigerian people and management context, CIPM.
- **Chartered Institute of Logistics and Transport Nigeria:** Nigerian logistics context, CILT Nigeria.
- **National Commission for Colleges of Education:** Nigerian education standards, NCCE.
- **National Agency for Food and Drug Administration and Control and Veterinary Council of Nigeria:** Nigerian agriculture, food, and veterinary boundaries, NAFDAC and VCN.
- **Nigerian Civil Aviation Authority and Nigerian Maritime Administration and Safety Agency:** aviation and maritime context, NCAA and NIMASA.
- **National Film and Video Censors Board:** Nigerian creative and media context, NFVCB.
- **National Lottery Regulatory Commission:** Nigerian gaming and lottery regulation, NLRC.
- **National Environmental Standards and Regulations Enforcement Agency:** Nigerian environmental regulation, NESREA.
- **Federal Ministry of Transportation:** Nigerian transportation context, Transportation Ministry.
- **Federal Ministry of Aviation and Aerospace Development:** Nigerian aviation and aerospace context, Aviation Ministry.
- **Choice Talents Nigeria game-developer listing:** one dated local role signal, LinkedIn listing.

Some source families appear under more than one Nigerian institutional name because they answer different questions. For example, a regulator may establish a practice boundary while a vacancy source establishes only a dated market signal. They must not be merged into one claim.

22. Record-level provenance: what must be retained

A complete record for a subject rule, programme, domain, subgroup, role, competency, tool, or credential should retain:

1. the canonical identity and any source aliases;
2. the source name and version;
3. the public source link, when available;
4. the page, section, table, or other locator;
5. the relevant source text or a faithful summary;
6. the evidence type: admission, curriculum, occupation, skill, tool, credential, Nigerian context, Nigerian market, remote, or global;
7. the relationship created from that evidence;
8. the rule applied;
9. the label class: exact, source-supported umbrella, or curated addition;
10. the confidence or review state;
11. the bridge or limitation a learner must see; and
12. the last review date.

This structure lets a reviewer trace a learner-facing sentence back through its relationship to the source. It also prevents a source link from being mistaken for proof of every field on the record.

23. What the product says and does not say

The product may say:

- “This programme matches the checked O'Level requirement.”
- “This programme matches the checked UTME requirement.”
- “This university offers this national programme.”
- “This curriculum teaches or introduces these skills.”
- “This domain is a Direct, Functional, or Adjacent relationship under the stated evidence rule.”
- “These are the first capabilities to build for this route.”
- “This source provides Nigerian context.”
- “This checked Nigerian source contains this market signal.”
- “This role has global evidence, but no direct Nigerian counterpart was found in the checked sources.”

The product must not say:

- that a subject match guarantees admission;
- that a programme guarantees employment;
- that a global role is automatically in Nigerian demand;
- that every tool is required;
- that every certification is a Nigerian licence;
- that a learner has mastered every skill listed in a curriculum; or
- that an unresolved source code has been corrected when it has not.

24. Closing explanation

Sifter is best understood as a chain of justified translations:

- examination language is translated into a checked programme possibility;
- a national programme name is translated into a university-specific offering;
- curriculum language is translated into plain-language skills taught and graduate capabilities;
- economic activity and occupational language are translated into learner-facing domains;
- occupational families are translated into supported subgroups and roles;
- source skills and professional standards are translated into competencies, tools, certifications, resources, and preparation;
- the evidence is translated into an ordered guidance route without hiding the full inventory.

At every translation, the source, rule, example, limitation, and learner meaning must remain clear. That is what makes the product explainable rather than a collection of unexplained labels or similarity scores.